AI201 (Fundamentals of Artificial Intelligence) **Term: First Semester 2026-27**
Instructor: Pros Naval (pcnaval@up.edu.ph) **Credit: 3 units**
Schedule/Venue: 6 to 9 pm on Thursdays / ERDT Room DCS

Course Description: Fundamental concepts in Artificial Intelligence. Problem formulation and knowledge representation for intelligent agents and their environments using symbolic and non-symbolic inference procedures. Supervised and unsupervised learning. Ethical considerations for AI.

Course Outcomes:
- describe the nature of artificial intelligence and its relationship with other fields of human endeavor;
- explain the notion of intelligent agents and how autonomous systems can be built from them;
- formulate problems that involve state-space search for performing heuristic search and adversarial search for game playing;
- formulate problems that involve combinatorial configuration constraints as constraint satisfaction problems and create solutions using constraint programming;
- formulate certain optimization problems as search, where candidate solutions are iteratively improved though evolutionary processes or collective behavior of populations, among others;
- demonstrate understanding on how a machine can learn from data using the three learning paradigms of supervised, unsupervised and reinforcement learning;
- describe the key application areas of AI;
- demonstrate understanding of key ethical issues related to the use of AI technologies, including their use in warfare, manipulation of public opinion, machine bias and risks, and responsibilities related to AI software deployment

Week	Date	Topic	Programming
1	Aug 13	What is Artificial Intelligence?	
2	Aug 20	Intelligent Agents	
3	Aug 27	Problem Solving by Search / Adversarial Search	Programming Assignment 1 Deadline: Sept 9
4	Sept 3	Constraint Satisfaction Problems	
5	Sept 10	Probabilistic Reasoning	Programming Assignment 2 Deadline: Sept 23
6	Sept 17	Supervised Learning I (Linear Classifiers)	
7	Sept 24	Supervised Learning II (Neural Networks)	Programming Assignment 3 Deadline: Oct 14
8	Oct 1	Exam 1 (Coverage: Weeks 1 to 7)	
9	Oct 8	Supervised Learning III (Support Vector Machines)	
10	Oct 15	Supervised Learning IV (Classifier Ensembles)	Programming Assignment 4 Deadline: Oct 28
11	Oct 22	Evolutionary Computation (Genetic Algorithm and Particle Swarm Optimization)	(Papers for Presentation)
12	Oct 29	Unsupervised Learning I (Principal Component Analysis)	Mini-Project Proposals
13	Nov 5	Unsupervised Learning II (Clustering)	(Mini-Project)
14	Nov 12	Reinforcement Learning (Q-Learning)	(Mini-Project)
15	Nov 19	Student Reports	(Mini-Project)
16	Nov 26	Exam 2 (Coverage: Weeks 9 to 14)	(Mini-Project)
17	Dec 3	Project Presentations	(Mini-Project)

Programming Language: Python
Course Requirements:
Recitation, Problems Sets, Quizzes, etc. ………......................... 5 %
Programming Assignments .... 40 %
Mini-Project .... 25 %
Reports .... 10 %
Exam ………………………………………………………….. 20 %
Class Policies:
1. Physical attendance required: **maximum of 3 absences** for the semester.
2. A student who intends to drop the course must do so officially. Last day for dropping is November 5, 2026.
3. Programming Assignments: must submit all Programming Assignments **at most once**.
4. All requirements are important. Missing one or more will result in an INC or failing grade.
5. Submission address for all requirements: submit2pcnaval@gmail.com
6. Examination requires **physical presence**. Make sure you can come on the day of exam.
7. Code of Conduct: see https://cswcd.upd.edu.ph/wp-content/uploads/2021/04/2012-Code-of-Student-Conduct-of-UP-Diliman.pdf